RESEARCH INTELLIGENCE & SCIENTIFIC AUDIT

SIMUPHARM.IO
EXPERIMENTAL VALIDATION ROADMAP

Quantitative Benchmarking Protocols, Reference Standards, and Statistical Defensibility Blueprints for Peer-Reviewed Publication

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Lead Auditor:	Scientific Validation & Pre-Publication Due Diligence Directorate
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This document defines the minimal, non-negotiable experimental and statistical validation required to elevate the existing in silico SimuPharm software baseline to peer-reviewed publication grade in indexed Q1/Q2 journals without modifying original project architecture.

1. Executive Mandate & The Empirical Gap

The scientific audit of the existing SimuPharm.io codebase established that while the client-side chromatographic engine (`hplc_simulator_script.js`) and multi-instrument modules contain sophisticated mathematical formulations, the project currently possesses **zero empirical validation against physical laboratory instruments**. In the automated test directory (`tests/test_simulations.py`), all 16 simulation tests are HTTP route smoke tests asserting status codes (`assert response.status_code in (200, 422)`) rather than verifying physical accuracy.

> **CRITICAL AUDIT FINDING:** An unbenchmarked simulator is considered an unvalidated mathematical hypothesis by peer-reviewed journals. Attempting to publish claims such as "*retention times are predicted within ±10–20%*" or "*accurate physical digital twin*" without the protocols outlined in this roadmap will trigger immediate rejection by hostile peer reviewers.

This roadmap establishes six targeted benchmarking protocols that utilize publicly accessible pharmacopoeial reference standards, certified reference materials (CRMs), and established chromatographic test mixtures. Executing these protocols requires **no modification of the SimuPharm codebase**; rather, it records outputs generated by the existing platform and statistically compares them to empirical ground truth.

2. Benchmark Protocol 1: Reversed-Phase HPLC Isocratic Kinetics & Van Deemter Validation

PROTOCOL SPECIFICATION: HPLC-ISO-01

Parameter	Specification
Scientific Objective	Quantify the accuracy of the Van Deemter plate height equation and isocratic retention factor prediction across variable linear flow velocities.
Core Hypothesis	Simulated plate heights (H_{sim}) will demonstrate a minimum (H_{min}) within ±15% of empirical column test data, and retention times will correlate with $R^2 \ge 0.95$ across 5 flow rates.
Test Analytes (Standard Mixture)	Tanaka Test Mixture / Standard Hydrophobic Mixture: 1. Uracil (unretained hold-up marker, t_0) 2. Acetophenone ($k' □pprox 1.5$) 3. Benzene ($k' □pprox 3.0$) 4. Toluene ($k' □pprox 4.5$) 5. Naphthalene ($k' □pprox 7.0$)
Reference Physical Column	Agilent Zorbax Eclipse Plus C18, $4.6 imes 150$ mm, $5.0\ \mu ext{m}$ particle diameter (d_p), porosity $□arepsilon □pprox 0.40$.
Independent Variables	1. Flow rate (F): 0.5, 0.8, 1.0, 1.5, 2.0 mL/min (linear velocity u: 0.83 to 3.32 mm/s). 2. Mobile phase composition: Methanol:Water 65:35 (v/v) and Acetonitrile:Water 50:50 (v/v). 3. Column temperature: Controlled at $30^\circ ext{C}$.
Dependent Variables	Retention time (t_R), retention factor ($k = (t_R - t_0)/t_0$), peak width at half-height ($W_{0.5}$), theoretical plates ($N = 5.54 (t_R / W_{0.5})^2$), plate height ($H = L / N$), system backpressure ($\Delta P$).
Repetitions & Controls	$N = 3$ physical replicates per condition; baseline noise recorded across blank injections.

Mathematical Evaluation Metrics

```
RMSE(t_R) = sqrt( (1/n) * sum_{i=1}^n (t_{R,sim,i} - t_{R,exp,i})^2 )
MAPE(t_R) = (100 / n) * sum_{i=1}^n |t_{R,sim,i} - t_{R,exp,i}| / t_{R,exp,i}
Two One-Sided Equivalence Test (TOST): H_0: |mu_diff| >= 0.15 * mu_exp vs H_1:
|mu_diff| < 0.15 * mu_exp (alpha = 0.05)
```

Acceptance Criteria

- **Retention Factor Correlation:** $R^2 \ge 0.95$ across all 5 test solutes.
- **Optimal Velocity (u_{opt}):** Modeled minimum plate height H_{min} occurs at 1.0 ± 0.3 mL/min, matching classical Van Deemter behavior.
- **Backpressure Concordance:** Modeled Darcy/Kozeny-Carman pressure drop within ±20% of physical manometer readouts across mobile phase viscosities.

3. Benchmark Protocol 2: Linear Solvent Strength (LSS) Gradient Elution & Band Compression Validation

PROTOCOL SPECIFICATION: HPLC-GRAD-02

Parameter	Specification
Scientific Objective	Validate the client-side numerical integration engine ($\Delta t = 0.01\ ext{min}$) in `hplc_simulator_script.js` for gradient retention time and peak compression predictions.
Core Hypothesis	The numerical solver $\int_0^{t_R} ↟rac{dt}{t_{R,inst}(\phi(t))} = 1$ correctly predicts gradient retention times with MAPE $\le 8.5\%$ and reproduces peak focusing factor compression within ±12% of empirical standards.
Model Pharmaceutical Mixture	Four-Component Active Pharmaceutical Ingredient (API) Formulation: 1. Paracetamol (polar analgesic, $\log P = 0.46$) 2. Caffeine (xanthine alkaloid, $\log P = -0.07$) 3. Aspirin (acetylsalicylic acid, $\log P = 1.19$) 4. Ibuprofen (lipophilic NSAID, $\log P = 3.97$)
Gradient Programs Tested	• **Linear Ramp A (Standard):** 10% to 90% B over 15 min ($5.33\%/ ext{min}$). • **Linear Ramp B (Shallow):** 10% to 70% B over 30 min ($2.00\%/ ext{min}$). • **Step Gradient C:** 10% B (2 min hold) → ramp to 60% B at 10 min → ramp to 90% B at 15 min.
Validation Ground Truth	Certified physical chromatographic dataset acquired on Waters Alliance 2695 with Waters 2996 PDA detector; column C18, $4.6 imes 250$ mm, flow rate 1.2 mL/min.

Expected Deliverable Plots & Tables for Manuscript

- **Figure 1:** Overlaid chromatograms of physical Waters Alliance acquisition vs. SimuPharm in silico simulation across Ramps A, B, and C.
- **Figure 2:** Residual parity plot ($t_{R,sim}$ vs. $t_{R,exp}$) with 95% prediction intervals and Bland-Altman limit of agreement lines.
- **Table 1:** Tabulated retention times, USP resolution (R_s), USP tailing (T), and capacity factors (k^*) comparing physical vs. simulated runs.

4. Benchmark Protocol 3: Gas Chromatography Golay Capillary Dispersion & Thermal Ramping

PROTOCOL SPECIFICATION: GC-GOLAY-03

Parameter	Specification
Scientific Objective	Assess the physical defensibility of Golay equation modeling, carrier gas diffusivity temperature dependence, and Kovats retention index interpolation.
Reference Mixture	Homologous n-Alkane Series (C_8 Octane through C_{16} Hexadecane) + BTEX mixture (Benzene, Toluene, Ethylbenzene, o-Xylene) in isooctane.
Reference Column	HP-5MS Capillary Column: Length $30\ ext{m}$, Inner Diameter $0.25\ ext{mm}$, Film Thickness $0.25\ \mu ext{m}$.
Carrier Gases	Helium (He, standard) vs. Nitrogen (N_2, low diffusivity) vs. Hydrogen (H_2, high diffusivity).
Oven Programs	• Isothermal: $100^\circ ext{C}$ constant. • Single Ramp: $50^\circ ext{C}$ (2 min hold) → $10^\circ ext{C}/ ext{min}$ to $250^\circ ext{C}$ (5 min hold). • Ballistic Ramp: $50^\circ ext{C}$ → $25^\circ ext{C}/ ext{min}$ to $300^\circ ext{C}$.

Methodological Boundary Disclosure Required

> The audit confirmed that `gc_script.js` line 853 computes retention using an empirical heuristic: $k \sim \exp((bp - T_{avg})/50)$. The validation paper must explicitly disclose this limitation and propose replacing the boiling-point heuristic with the thermodynamic relationship: $$\ln k = -rac{\Delta H_s^\circ}{RT} + rac{\Delta S_s^\circ}{R} + \ln \Phi$$

5. Benchmark Protocol 4: UV-Vis Spectrophotometric Linearity & Scipy Peak Prominence

PROTOCOL SPECIFICATION: UV-SPEC-04

Parameter	Specification
Scientific Objective	Validate Beer-Lambert linearity, molar absorptivity ($□arepsilon$) accuracy, and Scipy prominence peak detection limits.
Certified Standards	Potassium Dichromate ($K_2Cr_2O_7$) in $0.005\ ext{M}\ H_2SO_4$ (NIST Standard Reference Material SRM 935a) at concentrations: 20, 40, 60, 80, 100 mg/L.
Characteristic Peaks	Wavelengths: $235\ ext{nm}$, $257\ ext{nm}$, $313\ ext{nm}$, $350\ ext{nm}$.
Acceptance Criteria	1. Linearity: $R^2 \ge 0.9990$ across the calibration range 0.1 to $1.5\ ext{AU}$. 2. Absolute Absorptivity: Peak λ_{max} positions within $\pm 1.0\ ext{nm}$ of NIST SRM 935a values. 3. Automated Detection: Scipy `find_peaks` prominence threshold resolves all four absorption extrema without false positive shoulder artifact detection.

6. Benchmark Protocol 5: Pedagogical Fault-Injection Diagnostic Sensitivity Trial

PROTOCOL SPECIFICATION: PED-FAULT-05 (Human-Subjects Educational Trial)

Parameter	Specification
Educational Objective	Quantify the pedagogical efficacy of SimuPharm's fault injection engine (air bubble, column degradation, leaks) on undergraduate pharmacy students' troubleshooting competency.
Study Design	Controlled, randomized, crossover pre/post intervention study.
Participant Cohort	$N = 40$ third-year Bachelor of Pharmacy (B.Pharm) students enrolled in Pharmaceutical Analysis.
Arm Allocation	• **Cohort A ($N=20$):** Traditional lecture + ideal (error-free) simulator. • **Cohort B ($N=20$):** Traditional lecture + SimuPharm fault-injection active troubleshooting mode.
Assessment Instrument	Standardized 20-item diagnostic test presenting anomalous real-world chromatograms (split peaks, ghost peaks, drifting baselines, pressure drops).
Primary Outcome Metric	Hake's Normalized Learning Gain: `g = (Score_{post} - Score_{pre}) / (100 - Score_{pre})`
Statistical Hypothesis	$H_0: g_B \le g_A$ versus $H_1: g_B > g_A$ (one-tailed Welch's two-sample t-test, $\alpha = 0.05$, statistical power $1 - \beta = 0.85$).
Ethics & Compliance	Institutional Review Board (IRB) / Ethics Committee approval; informed written consent; anonymized performance metrics.

Significance for Top-Tier Publication

> **STRATEGIC ADVANTAGE:** Including this educational trial transforms the manuscript into an empirical human-subjects pedagogical study, which is the primary criterion for publication in the *Journal of Chemical Education* (ACS). This removes dependence on pure chromatographic novelty and guarantees an exceptionally strong scope fit!

7. Resource Allocation & 12-Week Execution Schedule

Phase & Weeks	Milestone Activities	Personnel Required	Equipment / Data Source	Deliverables
Weeks 1–2: Dataset Compilation	Extract certified retention times and spectra from NIST, SDBS, and Waters application libraries.	1 Research Assistant	Open-access databases (Zero Cost)	`benchmark_ground_truth.csv`
Weeks 3–5: In Silico Benchmarking	Execute SimuPharm simulators against reference parameters;	1 Lead Computational Scientist	Existing SimuPharm codebase	`sim_vs_exp_data_matrix.xlsx`

	record t_R, peak widths, pressures.			
Weeks 6–8: Physical Laboratory Runs	Run 15 validation injections of 4-component analgesic mixture on physical HPLC (Agilent/Waters).	1 Analytical Chemist	College analytical QC laboratory HPLC	Raw `.arw` / `.csv` physical detector files
Weeks 9–10: Educational Trial	Administer pre-test, simulator intervention, and post-test across $N=40$ B.Pharm students.	1 Faculty Instructor + 1 Observer	College computer laboratory	Student diagnostic score ledger
Weeks 11–12: Statistical Synthesis	Compute RMSE, MAPE, TOST equivalence intervals, and normalized learning gains (g). Generate figures.	Lead Author / Biostatistician	Python Scipy / Pandas / Matplotlib	Publication-ready Figures 1–6 & Tables 1–4

— END OF VALIDATION ROADMAP —